崑山科技大學外籍生大學部四年制機械工程課程科目表

課程名稱(講授時數－實習時數－學分數)

112.5.2 系課程委員會修訂通過
112.5.17 院課程委員會通過
112.7.20 校課程委員會通過

111 學年度入學適用

1st Academic Year		2nd Academic Year		3rd Academic Year		4th Academic Year	
1st Semester	2nd Semester	1st Semester	2nd Semester	1st Semester	2nd Semester	1st Semester	2nd Semester
General Courses (Total: 26 Cs)							
Foreign Language I (華語(一)) (3-0-3)	Foreign Language II (華語(二)) (3-0-3) Labor Education I (勞作教育(一)) (0-1-0)	Foreign Language III (華語(三)) (3-0-3) Labor Education II (勞作教育(二)) (0-1-0)	Foreign Language IV (華語(四)) (3-0-3) General Education I– Taiwan Culture (台灣文化) (2-0-2)	Foreign Language V (華語(五)) (3-0-3) General Education II– Modern Technology (現代科技) (2-0-2)	Foreign Language VI (華語(六)) (3-0-3) General Education III– Technique Writing (華語寫作技巧) (2-0-2)	General Education IV Foreign Language Seminar (華語文專題) (2-0-2)	
3-1-3	3-1-3	3-0-3	5-0-5	5-0-5	5-0-5	2-0-2	
Compulsory Courses 必修(Total: 62Cs)							
Calculus I (微積分(一)) (3-0-3) General Physics and Experiment I (普通物理與實驗(一)) (2-2-3) Chemistry (普通化學) (3-0-3) Engineering Graphics (圖學) (1-3-2) Shop Work I (工廠實習(一)) (0-3-1) Computer Programming (計算機程式) (1-3-2)	Calculus II (微積分(二)) (3-0-3) General Physics and Experiment II (普通物理與實驗(二)) (2-2-3) Shop Work II (工廠實習(二)) (0-3-1) Applied Mechanics 應用力學 (3-0-3) Engineering Materials (機械材料) (3-0-3) Computer Aided Drawing (電腦輔助繪圖) (1-2-2)	Engineering Mathematics I (工程數學(一)) (3-0-3) Thermodynamics (熱力學) (3-0-3) Mechanics of materials (材料力學) (3-0-3) Mechanisms (機構學) (3-0-3) Manufacturing Process (機械製造) (3-0-3) Electronics and Lab (電子學及實習) (2-2-3)	Engineering Mathematics II 工程數學(二) (3-0-3) Thermal Lab (熱工實驗) (0-3-1) Design of Mechanical Elements (機械元件設計) (3-0-3) Materials Lab (材料實驗) (0-3-1)	HandsOn I (實務專題(一)) (0-3-1) Numerical Methods and Lab (數值分析及實習) (2-2-3)	HandsOn II (實務專題(二)) (0-4-2)	HandsOn III (實務專題(三)) (0-3-1)	
10-11-14	12-7-15	17-2-18	6-6-8	2-5-4	0-4-2	0-3-1	0-0-0

Elective Courses 選修

(Based on every semester opening courses, the minimum requirement of total credits: 40 Cs)

1st Academic Year		2nd Academic Year		3rd Academic Year		4th Academic Year	
1st Semester	2nd Semester	1st Semester	2nd Semester	1st Semester	2nd Semester	1st Semester	2nd Semester
	Physical Fitness Experience (體適能體驗) (2-0-1)	Computer Engineering Applications and Lab (電腦工程應用及實習) (2-2-3) Pneumatics and Hydraulics Lab (氣液壓學及實習) (1-2-2) Non-traditional Machining (非傳統加工) (3-0-3) Soccer and Management (足球訓練與管理) (2-0-1) 磨潤分析 (3-0-3)	CAD/CAM (電腦輔助設計與製造) (2-2-3) Programmable Logic Controller (可程式控制原理與應用) (1-2-2) Photo- electronics (光電工程) (3-0-3) Surface Engineering Technology (表面工程技術) (3-0-3) Dynamics (動力學) (3-0-3) Introduction to Automatic Control (自動控制) (3-0-3) Introduction to Plastic Injection Molding (塑膠射出成型概論) (3-0-3) 機械信號分析 (3-0-3)	Computer Aided Engineering (電腦輔助工程分析) (2-2-3) Introduction to Sensors and Measurements (感測與量測概論) (2-0-2) Electro magnetism (電磁學) (3-0-3) Manufacturing Technology Analysis (製造技術分析) (3-0-3) Vibrations (振動學) (3-0-3) Heat Transfer (熱傳學) (3-0-3) Introduction to Computational Fluid Dynamics (計算流力概論) (3-0-3) Lubrication Analysis 磨潤分析 (3-0-3)	Computer Aided Rapid Prototyping (電腦輔助快速原型製作) (1-2-2) Fluid Mechanics (流體力學) (3-0-3) Heat Treatment (熱處理) (3-0-3) Introduction to Finite Element Method (有限元素法概論) (3-0-3) Refrigeration and Air-conditioning Technology（冷凍空調） (3-0-3) Fluid Mechanics Lab (流固力實驗) (0-3-1) Industrial Management (工業管理) (3-0-3) Automotive Engineering (汽車工程) (3-0-3) Mechanical Signal Analysis 機械信號分析 (3-0-3)	CAE in Mechanism and Dynamic (機構與動力分析及實習) (2-2-3) Mechatronic and Lab (機電整合及實習) (2-2-3) Reverse Engineering and Lab (逆向工程及實習) (2-2-3) Mechanism Design (機構設計) (3-0-3) Noise Control (噪音控制) (3-0-3) Introduction to Energy Engineering (能源工程概論) (3-0-3) M.E. Seminar I (機械工程專題討論(一)) (0-2-1) Industrial Design (工業產品設計) (3-0-3) Summer Off-Campus Internship (暑期校外實習) 0-*-2 Off-Campus Internship (校外實習) 0-*-2	Digital Image Processing and Lab (數位影像處理及實習) (2-2-3) Creative Design Methods (創意設計技法) (3-0-3) Introduction to Semi-Industry Equipments (資訊設備工程導論) (3-0-3) M.E. Seminar II (機械工程專題討論(二)) (0-2-1) Quality Control (品質管制) (2-0-2) Advanced Automotive Engineering (高等汽車工程) (3-0-3)

Minimum for Total Credits: 128 Cs

Note:

1. **(Le－Lb－Cs), Le: Lecture hours, Lb: Lab hours, Cs: Credits;**
2. **The university reserves the right to make changes to the courses offered without prior notice.**

The minimum requirement for a B.S. degree is to complete one-hundred-twenty-eight credits which are possibly included the exempt credits from the transfer university.

3. Summer Off-Campus Internship and Off-Campus Internship are both held in summer vacation. Summer Off-Campus Internship (2 credits) is granted for the first 160-hour intern practice while Off-Campus Internship (also 2 credits) is granted for the second 160-hour intern practice.